

## Communication

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 04 July 2026                            |
| <b>Team ID</b>       |                                         |
| <b>Project Name</b>  | Rising Waters – Flood Prediction System |
| <b>Maximum Marks</b> | 1 Mark                                  |

### Communication Plan:

Communication strategy used within the team and with stakeholders throughout the project lifecycle, including the later pass that added authentication and prediction history.

| S. N O | Communication Type                 | Frequency         | Channel / Tool                                              | Participants         | Purpose                                                                                   |
|--------|------------------------------------|-------------------|-------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------|
| 1      | <b>Progress Check-in</b>           | Per session       | Direct working sessions                                     | Lahari Sri Kotipalli | Track fixes, feature additions, and blockers as they came up                              |
| 2      | <b>Bug / Issue Discussion</b>      | As Needed         | In-context debugging during development                     | Lahari Sri Kotipalli | Resolve routing mismatches, field mismatches, and deployment config issues                |
| 3      | <b>Documentation Review</b>        | Per milestone     | README, demo script, and this documentation set             | Lahari Sri Kotipalli | Keep setup instructions and architecture notes accurate to the current code               |
| 4      | <b>Deployment Readiness Review</b> | Before deployment | Direct review of requirements.txt / Procfile / .env.example | Lahari Sri Kotipalli | Confirm the app (including new auth secrets) is ready to deploy on Render                 |
| 5      | <b>Security / Auth Review</b>      | Once, before demo | Direct review of forms.py, models.py, AUTH_SETUP.md         | Lahari Sri Kotipalli | Confirm CSRF protection, password hashing, and Google OAuth linking behave as expected    |
| 6      | <b>Final Demo Rehearsal</b>        | Once              | Video demo script walkthrough                               | Lahari Sri Kotipalli | Prepare a clear, complete project demonstration covering login through prediction history |

### Communication Challenges & Resolutions:

| S.N O | Challenge Faced                                             | Resolution / Action Taken                                                                          |
|-------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1     | Form fields didn't match what the prediction route expected | Audited app.py against predict.html and aligned both to the same 10 fields                         |
| 2     | GET /predict was rendering the wrong (duplicate) template   | Traced the routing logic and pointed it to the correct form template; removed the unused duplicate |

| S.N<br>O | Challenge Faced                                                                              | Resolution / Action Taken                                                                                                |
|----------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 3        | Uncertainty on Render deployment requirements                                                | Reviewed Render's Procfile/Gunicorn requirements and added the missing config                                            |
| 4        | Anonymous users could originally reach /predict directly                                     | Added Flask-Login with @login_required on /predict and /history, redirecting anonymous visitors to /login                |
| 5        | Risk of duplicate accounts when a user signed up with email first, then tried Google sign-in | google_callback() now matches on email and links the Google account to the existing user instead of creating a duplicate |